Polar Curves

Questions 1–90

2.51 — Question 1. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$ which are given in polar form:

a) $r = 2\sec\theta$ where $\theta \notin \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

b) $r = -6\operatorname{cosec}\theta$ where $\theta \notin \{0, \pi, 2\pi\}$

2.51 — Question 2. Sketch each of the following curves in polar form for $0 \leq \theta \leq 2\pi$. Label the values at which they cross the coordinate axes.

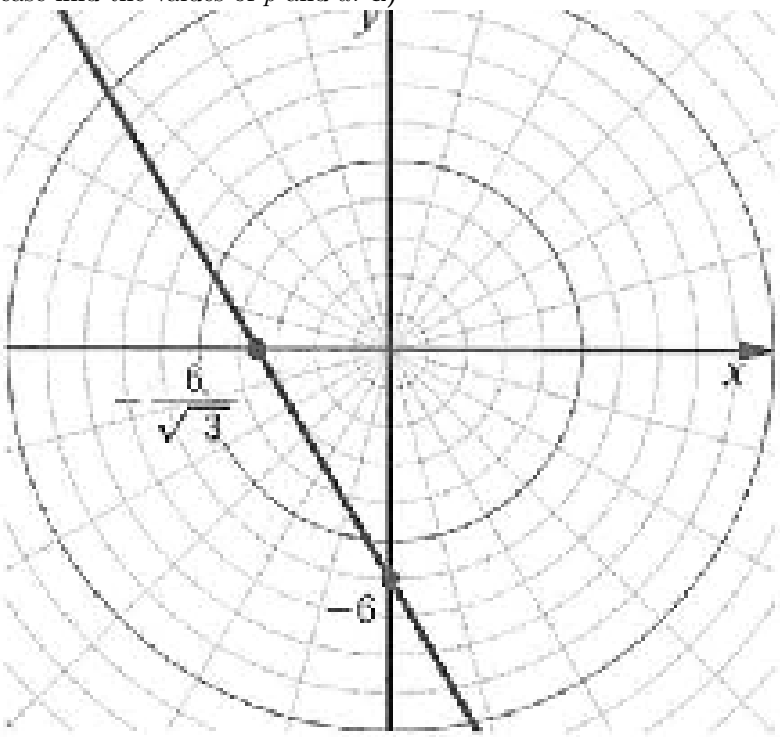
a) $r = 2\sec(\frac{\pi}{4} - \theta)$ where $\theta \notin \{\frac{\pi}{4}, \frac{5\pi}{4}\}$

b) $r = 2\operatorname{cosec}(\frac{\pi}{4} - \theta)$ where $\theta \notin \{\frac{3\pi}{4}, \frac{7\pi}{4}\}$

c) $r = -3\operatorname{cosec}(\frac{\pi}{6} - \theta)$ where $\theta \notin \{\frac{\pi}{6}, \frac{7\pi}{6}\}$

d) $r = -4\sec(\frac{5\pi}{6} - \theta)$ where $\theta \notin \{\frac{\pi}{3}, \frac{4\pi}{3}\}$

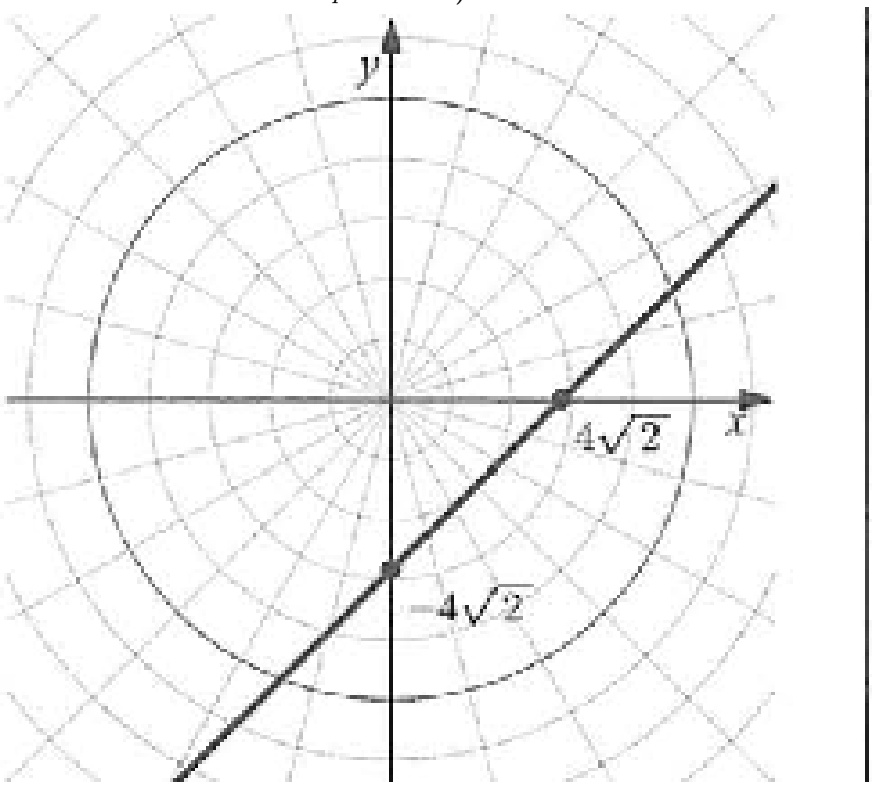
2.51 — Question 3. The following curves are of the form $r = p\sec(a-\theta)$ for integer p, $0 < a < \pi$ and $0 \leq \theta \leq 2\pi$. In each case find the values of p and a. **a)**



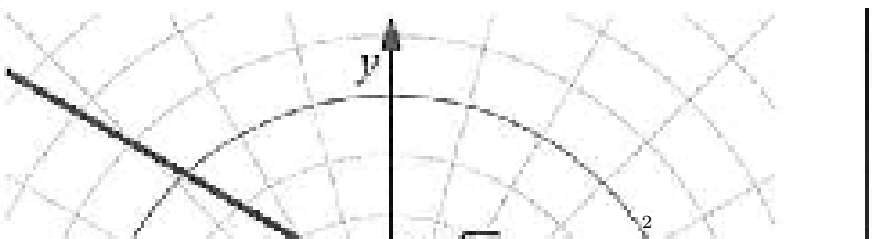
b)



2.51 — Question 4. The following curves are of the form $r = p\,\mathrm{cosec}(a-\theta)$ for integer p, $0 < a < \pi$ and $0 \leq \theta \leq 2\pi$. In each case find the values of p and a. **a)**



b)



2.51 — Question 5. Sketch each of the following, given that k is a positive constant, and $0 < \alpha < \frac{\pi}{2}$.

a) $r = k\sec(\alpha - \theta)$

b) $r = k\,\mathrm{cosec}(\alpha - \theta)$

2.51 — Question 6. Sketch $r^2 = 16\cos 2\theta$ for each of the following restrictions:

a) $0 \leq \theta \leq \frac{\pi}{4}$

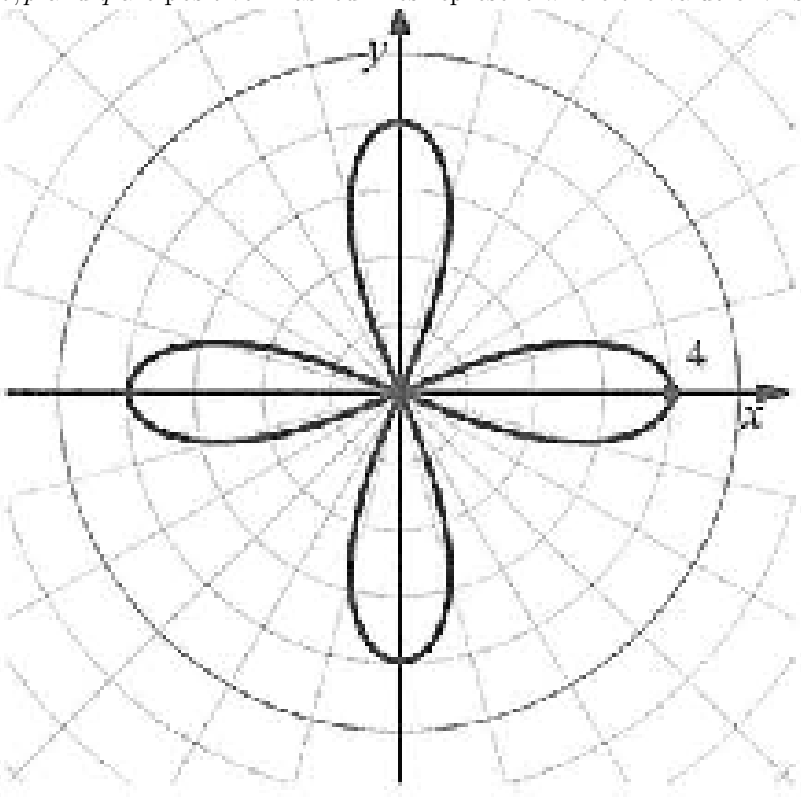
b) $\frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}$

c) $\frac{7\pi}{4} \leq \theta \leq 2\pi$

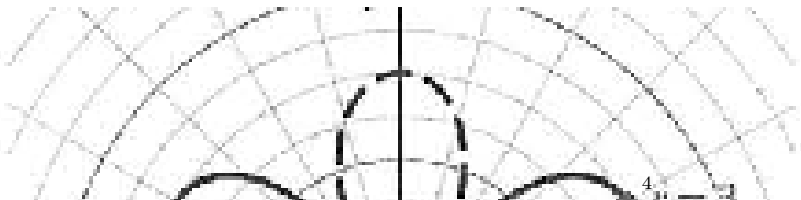
2.51 — Question 7. Sketch each of the following curves which are given in polar form. You may assume that θ is restricted to only values for which the curve exists. In each case label a point which lies furthest from the pole with its value of r:

a) $r^2 = 25\sin 2\theta$
b) $r^2 = 10\cos 3\theta$
c) $r^2 = 12\sin 4\theta$
d) $r^2 = 9\sin 6\theta$
e) $r^2 = 14\cos 5\theta$
f) $r^2 = 7\sin 5\theta$

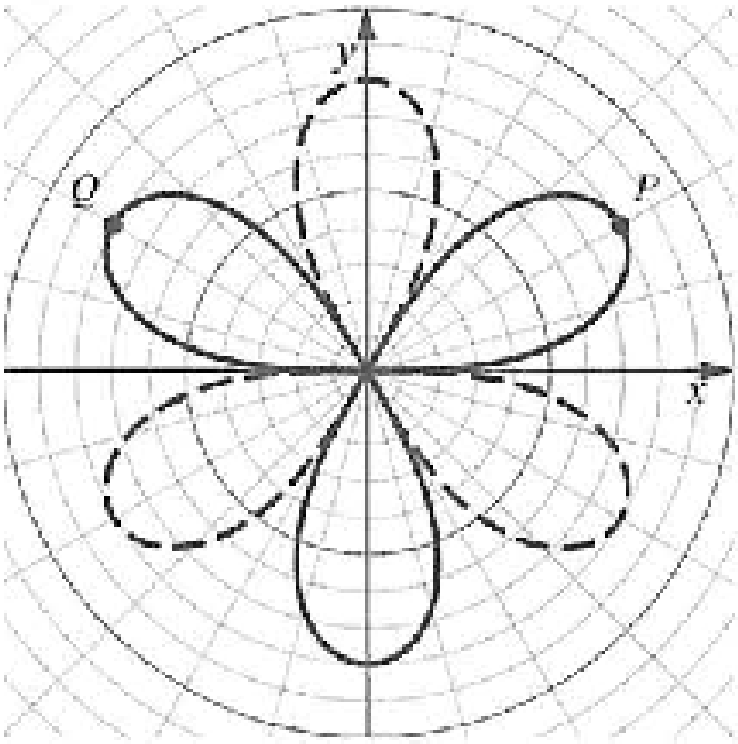
2.51 — Question 8. Find the polar curve which are equations of the form $r^2 = a\cos p\theta$ or $r^2 = b\sin q\theta$ where a, b, p and q are positive. Dashed lines represent where the value of r is negative. **a)**



b)



2.51 — Question 9. The curve C has polar equation $r^2 = 64\sin 3\theta$ and is shown on the graph below.



a) Find the Cartesian coordinates of the point P.
b) Find the Cartesian coordinates of the point Q.
c) Find the points of intersection of the curve C with a second curve given by $r = \sqrt{32}$ for $-\pi < \theta \leq \pi$ giving your

2.51 — Question 10. The curve C_1 is defined by $r^2 = 16\cos 4\theta$ for $0 < \theta < \pi$.
a) Determine the values of θ in the interval $0 < \theta < \pi$ for which the curve exists.
A second curve C_2 is given by $r = 4\cos 4\theta$ for $0 < \theta < 2\pi$.
b) Describe two similarities and two differences between the curves C_1 and C_2.

2.52 — Question 1. Write the following polar equations in Cartesian form, giving your answers in the form $ax + by = k$ where a, b and k are integers or surds to be found:

a) $r = \sec\left(\theta - \frac{\pi}{3}\right)$
b) $r = \sec\left(\theta + \frac{\pi}{6}\right)$
c) $r = \operatorname{cosec}\left(\theta + \frac{\pi}{4}\right)$
d) $r = 4\operatorname{cosec}\left(\theta - \frac{\pi}{6}\right)$

2.52 — Question 2. Find the Cartesian equation for each of the following curves:

a) $r = 3 + \cos 2\theta$
b) $r^2 = \sin 2\theta$
c) $r^2 = 3\sec 2\theta$
d) $r^2 = 9\operatorname{cosec} 2\theta$
e) $r = 6\cot\theta\operatorname{cosec}\theta$

2.52 — Question 3. The Cartesian equation

$$(x^2 + y^2)^2 = x^2 - y^2$$

when written in Polar form is given by $r^2 = \mathrm{f}(\theta)$. Find $\mathrm{f}(\theta)$ in its simplest form.

2.52 — Question 4. The Cartesian equation $y = \frac{1}{x}$ when written in Polar form is given by $r^2 = \mathrm{f}(\theta)$. Find $\mathrm{f}(\theta)$ in its simplest form.

2.52 — Question 5. It is known that the complex number z satisfies

$$|z - 6 + 8i| = 10$$

a) Sketch the locus of z on an Argand diagram.
b) Show that this locus of points can be represented by the polar curve

$$r = 12\cos\theta - 16\sin\theta$$

2.52 — Question 6.
a) Show on an Argand diagram the locus of points by the values of z satisfying

$$|z - 4 - 3i| = 5$$

b) Show that this locus of points can be represented by the polar curve

$$r = 8\cos\theta + 6\sin\theta$$

2.52 — Question 7. It is known that the complex number z satisfies

$$|z-1| = |z-\sqrt{3}i|$$

a) Sketch the locus of z on an Argand diagram.
b) Show that this locus of points can be represented by the polar curve

$$r = \frac{1}{2}\operatorname{cosec}\left(\theta - \frac{\pi}{6}\right)$$

2.52 — Question 8.

a) Sketch the polar curve whose polar equation is $r = 1 + 2\cos 2\theta$ for $0 \leq \theta \leq 2\pi$, demonstrating any negative parts using a dashed line.
b) Prove that the values of θ at the points of intersection of the curve $r = 1 + 2\cos 2\theta$ with the line $r = \sqrt{3}\sec\theta$ are satisfied by

$$4\cos^3\theta - \cos\theta - \sqrt{3} = 0$$

Prerequisite Knowledge for Q9: Hyperbolic Functions and Hyperbolic Identities

2.52 — Question 9. A curve C is given by

$$r = \sqrt{1 + \sinh^2 \theta}$$

A line L, given by $y = x$, intersects the curve C
Find the first two points of intersection between the line and the curve, giving your answers in exact form.

2.52 — Question 10. A curve C has Cartesian equation

$$x^3 - y^3 = 3xy \text{ where } x \leq 0 \text{ and } y \geq 0$$

The equation of the curve C in Polar form is given by $r = \mathrm{f}(\theta)$ where $\frac{\pi}{2} \leq \theta \leq \pi$

a) Find $\mathrm{f}(\theta)$ giving your answers in terms of $\sin\theta$ and $\cos\theta$
b) Find an expression for $\mathrm{f}\left(\frac{3\pi}{2} - \theta\right)$
c) Hence find the line of symmetry of C in its Cartesian form.
d) Show that C has a loop.

2.52 — Question 11. A curve C is given by

$$13x^2 - 6\sqrt{3}xy + 7y^2 = 1$$

a) Show that

$$13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta = 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta$$

b) Without using Calculus, find the greatest and least distances of points on the curve from the origin.

2.53 — Question 1. Find each of the following:

a) $\displaystyle\int (2\cos 3x)^2\,dx$

b) $\displaystyle\int (2\sin 3x)^2\,dx$

c) $\displaystyle\int (1+\cos x)^2\,dx$

d) $\displaystyle\int (1-\sin x)^2\,dx$

e) $\displaystyle\int \sin^2\left(\frac{x}{6}\right)\,dx$

f) $\displaystyle\int \left(2+\cos\frac{x}{2}\right)^2\,dx$

2.53 — Question 2. Find the exact area enclosed by each of the following curves:

a) $r = 4 + 3\cos\theta$ for $0 \leq \theta \leq 2\pi$

b) $r = 5 + 2\sin\theta$ for $0 \leq \theta \leq 2\pi$

c) $r = a(1 - \cos\theta)$ for $0 \leq \theta \leq 2\pi$, for positive constant a

d) $r = a(1 - \sin\theta)$ for $0 \leq \theta \leq 2\pi$, for positive constant a

2.53 — Question 3. In each of the following cases, use polar integration to find the area enclosed by the curve given, where a is a positive constant:

a) $r = a\cos\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

b) $r = a\sin\theta$ for $0 \leq \theta \leq \pi$

c) $r = a$

2.53 — Question 4. In each of the following cases, use polar integration to find the area bounded by the curve and the initial line, where a is a known positive constant:

a) $r = ae^{\theta}$ for $0 \leq \theta \leq \pi$

b) $r = a\theta$ for $0 \leq \theta \leq \pi$

2.53 — Question 5. Sketch $r = 2\sin 3\theta$ for $0 \leq \theta \leq \frac{\pi}{3}$.

b) On your diagram, shade the region for which $r = 2\sin 3\theta$ and $\theta \leq \frac{\pi}{6}$.

c) Find the exact area of the region you have shaded in part (b).

2.53 — Question 6. Show that the curve $r = a\sin^2\theta$ for $0 \leq \theta \leq \pi$ for positive constant a consists of a single closed loop through the pole.

a) Sketch the curve $r = a\sin^2\theta$ for $0 \leq \theta \leq \pi$, where a is a positive constant.

b) Find, in terms of a, the exact area of the region enclosed by the curve.

2.53 — Question 7. A student is attempting to find the area enclosed by the curve C given by $r^2 = 6\cos 3\theta$. Their working is shown below:

$$A = \frac{1}{2}\int_{-\pi}^{\pi} r^2\, d\theta = \frac{1}{2}\int_{-\pi}^{\pi} 6\cos 3\theta\, d\theta = \frac{1}{2}[2\sin 3\theta]_{-\pi}^{\pi} = 0.$$

a) Sketch the curve C.

b) Explain what the student has done wrong.

c) Find the area enclosed by one petal of the curve.

d) Hence, state the total area enclosed by the curve $r^2 = 6\cos 3\theta$.

2.53 — Question 8. Find the area enclosed by each of the following curves, giving your answer in exact form in terms of a:

a) $r^2 = a\sin 2\theta$
b) $r^2 = a\cos 2\theta$
c) $r^2 = a\sin 3\theta$
d) $r^2 = a\cos 3\theta$

2.53 — Question 9. Find the area enclosed by each of the following curves, giving your answer in exact form:

a) $r = 2\cos 3\theta$
b) $r = 3\sin 2\theta$
c) $r = a\cos 4\theta$ for positive constant a
d) $r = a\sin 5\theta$ for positive constant a

2.53 — Question 10. Find the area enclosed by the curve $r = \frac{1}{\sqrt{\theta}}$, the half-lines $\theta = \frac{\pi}{6}$ and $\theta = \frac{11\pi}{6}$, and the initial line.

2.53 — Question 11. Find the area of the inside loop enclosed by each of the following curves, giving your answer in exact form:

a) $r = 1 + 2\cos\theta$ for $0 \leq \theta < 2\pi$

b) $r = 2 - 4\sin\theta$ for $0 \leq \theta < 2\pi$

2.53 — Question 12. Show that the area enclosed by the curve $r = a(p + q\sin\theta)$ for $0 \leq \theta < 2\pi$, where $p \geq q$ and a is a positive constant, is given by $\frac{1}{2}a^2(2p^2 + q^2)\pi$ units^2.

2.53 — Question 13. For each curve, find the ratio of the area bounded by the curve above the initial line to the area bounded below it:

a) $r = 2(1 + \cos\theta)$ for $0 \leq \theta < 2\pi$

b) $r = 2(1 + \sin\theta)$ for $0 \leq \theta < 2\pi$

2.53 — Question 14. A hole for a pond has been dug with vertical sides and even depth 0.8 metres. The horizontal cross-section is modelled by the curve with equation

$$r = \frac{2}{5}(4 + \sin 4\theta), \qquad 0 \leq \theta \leq 2\pi,$$

where r is measured in metres.

a) Find the volume of water it will take to completely fill the pond, giving your answer in exact form.

b) State a modelling assumption that you have made.

2.53 — Question 15. A large paddling pool, with depth 0.4 metres and horizontal cross-section modelled by the curve with equation

$$r = 5 + \cos 4\theta, \qquad 0 \leq \theta < 2\pi,$$

where r is measured in metres. The paddling pool is initially empty, then the water is turned on and pours into the paddling pool at a rate of 40 litres per minute.

a) Determine how long it will take to completely fill the paddling pool, giving your answer to the nearest minute.

b) State a limitation of the model.

2.53 — Question 16. A curve is given by $r = ae^{k\theta}$ for $\theta \geq 0$, where a and k are positive constants. Find the area enclosed by the curve and the lines:

a) $\theta = 0$ and $\theta = \frac{\pi}{4}$
b) $\theta = 2\pi$ and $\theta = \frac{9\pi}{4}$
c) $\theta = 4\pi$ and $\theta = \frac{17\pi}{4}$
d) Hence show that the successive areas between the curve and the lines $\theta = 2n\pi$ and $\theta = 2n\pi + \frac{\pi}{4}$ increase in a geometric sequence. State the common ratio.

2.53 — Question 17. A curve is given by $r = 1 + \sinh\theta$ for $0 \leq \theta \leq \pi$ and consists of a single closed loop through the pole. Find the area enclosed by the curve and the initial line, giving your answer to three significant figures.

2.53 — Question 18. A curve is given by $r = \sqrt{3}\sin\theta$ for $0 \leq \theta \leq k$ for constant k.

a) State the smallest possible value of k for it to form a single loop.

b) Find the greatest value of r.

c) Hence sketch the curve.

d) Find the area enclosed by the curve.

2.53 — Question 19. A curve C_1 is given by $r = \sqrt{16\sin 2\theta}$ for $0 \leq \theta \leq k$ for constant k.

a) State the smallest possible value of k for it to form a single loop.

b) Find the area enclosed by C_1 for your value of k.

P is the point on the curve such that r is a maximum.

c) Find the polar coordinates (r, θ) of P.

The matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ represents the transformation T. T maps C_1 onto a curve C_2.

d) Show that the point P after it has been mapped by transformation T has an argument that satisfies $\tan\theta = \frac{2}{4}$.

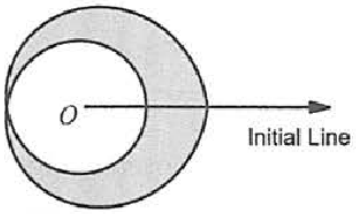
e) Find the area enclosed by C_2.

2.53 — Question 20. A curve is given by polar equation $r = 2 + 4\cos\theta$ for $0 \leq \theta < 2\pi$.

a) Sketch the curve.

b) Show that the area of the region inside the curve, but not inside the loop, is $(4\pi + 12\sqrt{3})$ units^2.

2.53 — Question 21. A curve is given by polar equation $r = 2 - \sin\left(\frac{\theta}{4}\right)$ for $0 \leq \theta \leq 4\pi$. Some regions enclosed by the curve as shown in the diagram have been shaded. Show that the area of the shaded region is $(-18 + 16\sqrt{2})$ units^2.



2.53 — Question 22.

a) Given that $z = \cos\theta + i\sin\theta$, express $z^n + \frac{1}{z^n}$ in terms of trigonometric functions of $n\theta$.

b) By considering $\left(z + \frac{1}{z}\right)^4$, show that

$$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}.$$

c) Sketch the curve with polar equation $r = a\cos^2\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

d) Find, in terms of a, the exact area of the region enclosed by the curve.

2.53 — Question 23. A curve is given by $(x^2 + y^2)^3 = y^4$.

a) Find the polar equation of the curve.

b) Sketch the curve.

c) Find the area enclosed by the curve.

2.53 — Question 24. A curve is given by $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ for constant a.

a) Find the polar equation of the curve.

b) Sketch the curve.

c) Find, in terms of a, the exact area of two of the loops.

2.53 — Question 25. A curve is given by $y^2 = a^2(x - y)^{4/3} - x^2$ for constant a.

a) Find the polar equation of the curve, giving your answer in the form $r^2 = f(\theta)$.

b) Show that $y = -x$ is a line of symmetry.

c) Sketch the curve.

d) Find, in terms of a, the exact area of two of the loops.

2.53 — Question 26. A curve is given by

$$(x^2 + y^2)^2 + ax(3y^2 - x^2) = 0$$

for positive constant a.

a) Given that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, show that the polar equation of the curve is $r = a\cos 3\theta$.

b) Sketch the curve.

c) Find, in terms of a, the exact area of three of the loops.

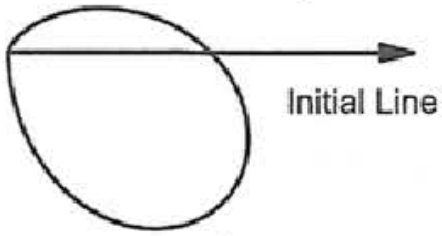
2.53 — Question 27. A curve has polar equation $r = 3\cos\theta + \sin\theta$ for $0 \leq \theta \leq \pi$.

a) Find the area enclosed by the curve.

b) Find the Cartesian coordinates of the point that is furthest away from the pole.

c) Show that the curve is a circle whose radius and centre should be specified.

2.53 — Question 28. The diagram shows the loop of the curve given by $r = 2(1 - \sin\theta)\cos\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Show that the area enclosed by the loop is $\frac{5\pi}{4}$ units^2.



2.53 — Question 29.

a) Sketch the curve $r = a\sec\theta$ for positive constant a.

b) Use polar integration to find the area enclosed by $r = a\sec\theta$, the half-line $\theta = k$ for $0 < k < \frac{\pi}{2}$, and the initial line. Give your answer in terms of a and $\tan k$.

2.53 — Question 30. Show that the area of the region enclosed by the curve given by

$$r = a\sec^2\left(\frac{\theta}{2}\right) \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2},$$

for positive constant a, the initial line and the line $\theta = \frac{\pi}{4}$ is

$$a^2\left(\tan\frac{\pi}{8} + \frac{1}{3}\tan^3\frac{\pi}{8}\right) \text{ units}^2.$$

2.53 — Question 31. A curve has polar equation

$$r = 2\sqrt{\cos\theta}\, e^{\frac{1}{3}\sin\theta}, \qquad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

a) Find the exact area enclosed by the curve.

b) Show that the greatest value of r on the curve is $2\sqrt{\frac{\sqrt{3}}{2}}e^{1/6}$.

2.53 — Question 32. A polar curve is given by $r = \frac{12}{k} + k(2 + \sin\theta)$ for $0 \leq \theta \leq 2\pi$, for positive constant k.

a) Show that the area enclosed by the curve is $\left(\frac{9}{2}k^2 + 48 + \frac{144}{k^2}\right)\pi$ units^2.

b) Find the exact value of k such that the area is minimised.

Compound Areas

2.53 — Question 33. Two curves C_1 and C_2 are defined by

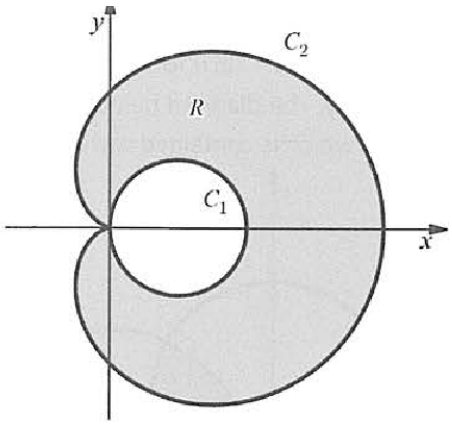
$$r_1 = \cos\theta \quad \text{for } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = a(1 + \cos\theta) \quad \text{for } 0 \leq \theta \leq 2\pi,$$

respectively, where a is a positive constant.

a) Write down the polar coordinates (r, θ) of the points where curve C_1 crosses the initial line.

b) Explain why both C_1 and C_2 are symmetrical about the initial line.

c) Find the area R which lies inside C_2 but outside C_1, giving your answer in terms of a.



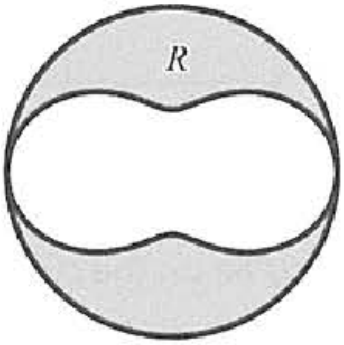
2.53 — Question 34. A design of a logo can be formed by the polar curve with equation $r = 2.1 + a\cos(2\theta)$ for $0 \leq \theta \leq 2\pi$, where a is a constant, and a circle which meets the curve twice and is centred around the pole. The logo is designed to be 6 units across.

a) State the value of a.

b) Find the exact area of the shaded region of the logo.

The logo is to be painted on the wall of a building so that it is 12 m wide and requires two coats of paint, and a tin of paint covers 40 m^2.

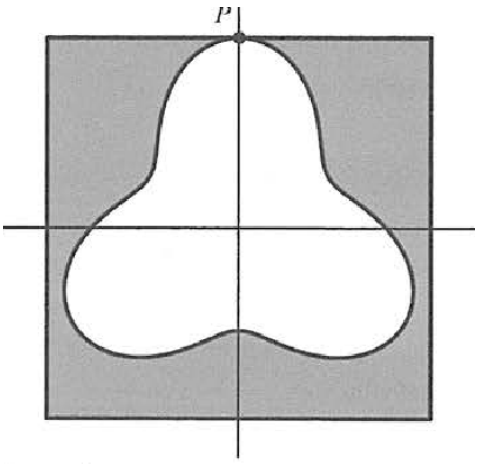
c) Determine how many tins of paint will be needed to paint the logo.



2.53 — Question 35. An artist is designing a decorative panel by cutting a square piece of wood with side length 2.6 m. The curve to be removed is defined by polar equation $r = 1 + a\sin 3\theta$ for $0 \leq \theta \leq 2\pi$ and $-1 < a < 1$, and is positioned so it is symmetric and touches the top centre of the square, labelled as point P in the diagram below.

a) Show that $a = -0.3$.

b) Find the area of the remaining wood, giving your answer in m^2 to three significant figures.



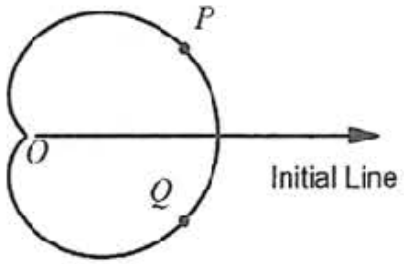
2.53 — Question 36. The curve C shown in the diagram below is given by the polar equation $r = a + b\cos\theta$ for $-\pi \leq \theta \leq \pi$, where a and b are positive constants and $a > b$. Points P and Q lie on the curve C such that OPQ is an equilateral triangle which is symmetrical across the initial line.

a) Find the polar equation of the line passing through both P and Q in terms of a and b.

Given the point P occurs when $r = 2$ and $b = 1$:

b) State the exact value of a.

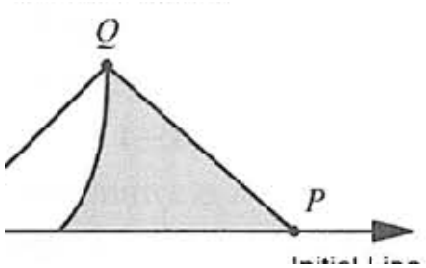
c) Show that the area of the equilateral triangle is approximately 30.9% of the area enclosed by the curve.



2.53 — Question 37. The diagram shows curve C given by polar equation $r = 1 + \tan\theta$ for $0 \leq \theta \leq \frac{\pi}{4}$, the line segment OQ given by $\theta = \frac{\pi}{4}$ and the point P which has coordinates $(3, 0)$.

a) Find the length of line OQ.

b) Show that the exact area of the shaded region is $\frac{1}{2}(3\sqrt{2} - 1 - \ln 2)$.



Intersecting Polar Regions

2.53 — Question 38. Two curves C_1 and C_2 are defined by

$$r_1 = 1 + \sin\theta \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 3\sin\theta \quad \text{for } 0 \leq \theta < \pi,$$

respectively.

a) Sketch C_1 and C_2 on the same axes.
b) Find the point(s) of intersection(s) of the curves.
c) Explain whether or not the curves intersect at the pole.

2.53 — Question 39. Two curves C_1 and C_2 are defined by

$$r_1 = 2(1 - \cos\theta) \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 2\cos\theta \quad \text{for } -\frac{\pi}{2} \leq \theta < \frac{\pi}{2},$$

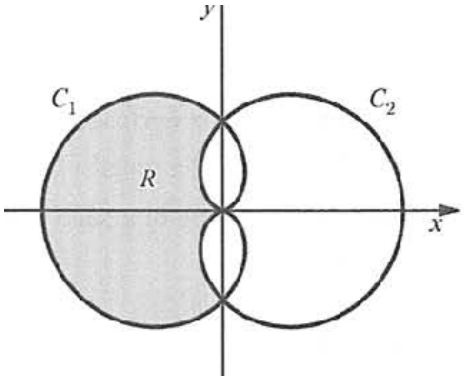
respectively.

a) Sketch C_1 and C_2 on the same axes.
b) Find the point(s) of intersection(s) of the curves.
c) Explain whether or not the curves intersect at the pole.

2.53 — Question 40. Two curves C_1 and C_2 are defined by

$$r_1 = 1 - \cos\theta \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 1 + \cos\theta \quad \text{for } 0 \leq \theta < 2\pi,$$

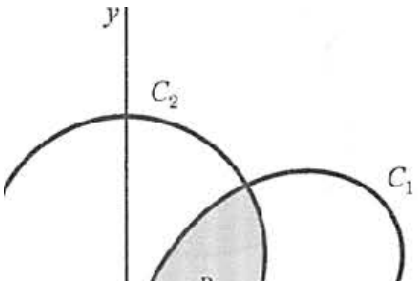
respectively. The diagram below shows the region R which lies inside C_1 but outside C_2. Determine the exact area of R.



2.53 — Question 41. Two curves C_1 and C_2 are defined by

$$r_1 = \cos\theta\sqrt{2\sin\theta} \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = 3^{-1/4}\sin\theta \quad \text{for } 0 \leq \theta \leq \pi,$$

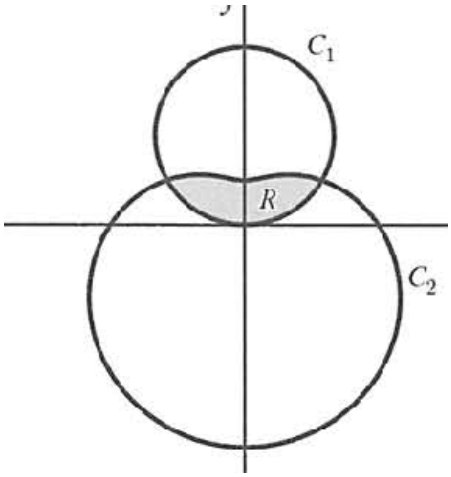
respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.



2.53 — Question 42. Two curves C_1 and C_2 are defined by

$$r_1 = 4\sin\theta \quad \text{for } 0 \leq \theta < \pi, \qquad r_2 = 3 - 2\sin\theta \quad \text{for } 0 \leq \theta < 2\pi,$$

respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.

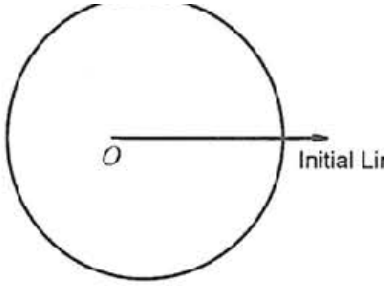


2.53 — Question 43. The line L has polar equation $r = \frac{7\sqrt{3}}{4}\operatorname{cosec}\theta$ for $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$ and the curve C has polar equation $r = 4 + \cos\theta$ for $-\pi \leq \theta \leq \pi$.

a) Verify that the two points of intersection occur when $\theta = \frac{2\pi}{3}$ and $\theta \approx 0.68817$.

b) On a copy of the diagram, add line L. The region R is the set of points which lie above the initial line such that $r < \frac{7\sqrt{3}}{4}\operatorname{cosec}\theta$ and $r < 4 + \cos\theta$.

c) Find the area of the region R, giving your answer to two significant figures.



2.53 — Question 44. For each of the following pairs of curves, C_1 and C_2, find the area of the region that lies inside C_1 but outside C_2.

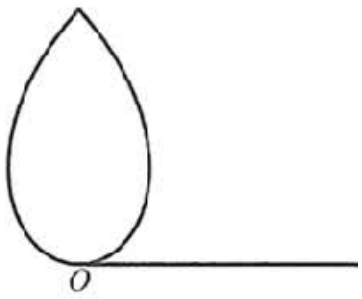
a) $r_1 = 3 + 2\sin\theta$ for $0 \leq \theta < 2\pi$ and $r_2 = 2$

b) $r_1 = 4 - 2\cos\theta$ for $0 \leq \theta \leq 2\pi$ and $r_2 = 6\cos\theta$ for $0 \leq \theta \leq \pi$

2.53 — Question 45. The curve C is given by $r = 2\sinh\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. The diagram below shows the curve C.

a) On a copy of the diagram add the curve given by $r = 2$.

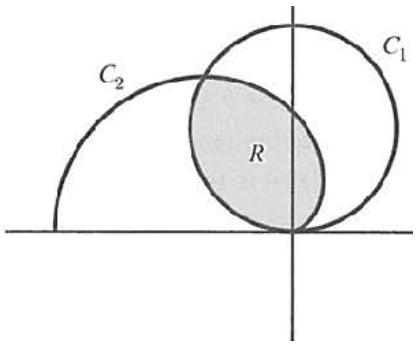
b) Show that the exact area of the region which lies inside both of the curves is $2\sqrt{2} + 2\pi - 6\ln(1 + \sqrt{2})$ units^2.



2.53 — Question 46. Two curves C_1 and C_2 are defined by

$$r_1 = 2\sin\theta \quad \text{for } 0 \leq \theta \leq \pi, \qquad r_2 = \frac{2\sqrt{3}}{3}(1-\cos\theta) \quad \text{for } 0 \leq \theta \leq \pi,$$

respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.



2.53 — Question 47. Two curves C_1 and C_2 are defined by

$$r_1 = 2\cos\theta \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = 1+\sin\theta \quad \text{for } 0 \leq \theta \leq \pi,$$

respectively. Show that an expression for the area of the region enclosed by C_1, C_2 and the initial line is $p\pi + q + r\phi$ units^2, where p, q and r are constants to be determined and $\sin\phi = \frac{3}{5}$.

2.54 — Question 1. For each of the following equations which are given in polar form, find y and x in terms of $\sin\theta$ and $\cos\theta$:

a) $r = 4\cos\theta$
b) $r = 8\sin\theta$
c) $r = 3(1 + \cos\theta)$
d) $r = 6(1 - \sin\theta)$
e) $r = 5\sin 2\theta$

2.54 — Question 2. For each of the following, find $\dfrac{dy}{d\theta}$ giving your answer in terms of $\sin\theta$ and $\cos\theta$:

a) $y = 3\cos\theta + 3\cos^2\theta$
b) $y = 5\sin\theta - 5\sin^2\theta$
c) $y = 7\cos\theta - 7\cos^2\theta$
d) $y = -2\sin\theta - 2\sin^2\theta$
e) $y = 8\cos^3\theta - 4\cos\theta$
f) $y = 6\sin^2\theta\cos\theta$

2.54 — Question 3. A polar curve is given by $r = 2(1 + \sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$.

a) Show that $\dfrac{dy}{d\theta} = 2\cos\theta + 2\sin 2\theta$.

b) Show that $\dfrac{dx}{d\theta} = -2\sin\theta + 2\cos 2\theta$.

c) Determine the values of θ for which $\dfrac{dy}{d\theta} = 0$.

d) Determine the values of θ for which $\dfrac{dx}{d\theta} = 0$.

e) Hence, without referring to the graph of $r = 2(1 + \sin\theta)$, identify where the curve has a cusp point.

2.54 — Question 4. Each of the following polar curves are defined for $r > 0$ and $-\pi < \theta \leq \pi$. In each case, find the equation of the tangent lines to the curve which are parallel to the initial line $\theta = 0$:

a) $r = 2(1 + \cos\theta)$

b) $r = 3(1 - \sin\theta)$

c) $r = 4\sin 2\theta$

d) $r = a(1 + \sin\theta)$ for constant a

e) $r = a(1 - \cos\theta)$ for constant a

f) $r = a\cos 2\theta$ for constant a

g) $r = e^{\theta}$

2.54 — Question 5. Each of the following polar curves are defined for $r > 0$ and $-\pi < \theta \leq \pi$. In each case, find the equation of the tangent lines to the curve which are perpendicular to the initial line $\theta = 0$:

a) $r = 4 - 4\cos\theta$
b) $r = 2 + 2\sin\theta$
c) $r = 2\cos 2\theta$
d) $r = a - a\sin\theta$ for constant a
e) $r = a + a\cos\theta$ for constant a
f) $r = a\sin 2\theta$ for constant a
g) $r = e^{\theta}$
h) $r = 1 + \tan\theta$

2.54 — Question 6. A polar curve is given by $r = 2 + \sqrt{3}\cos\theta$, $0 \leq \theta < 2\pi$. A point P where $0 < \phi < \dfrac{\pi}{2}$ is on the curve, and the tangent to the curve is parallel to the initial line.

a) Show that $\cos\phi = \dfrac{-1 + \sqrt{7}}{2\sqrt{3}}$.
b) Hence, find the value of r at the point P.

2.54 — Question 7. A polar curve is given by $r = e^{2\theta}$, $0 \leq \theta \leq \pi$.

a) Sketch the curve.

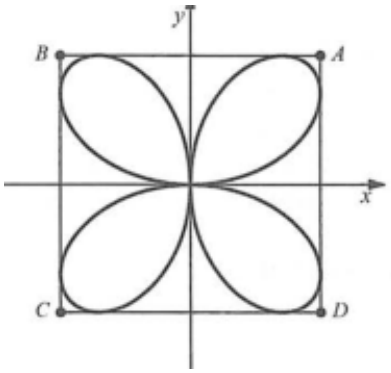
b) State the distance between the pole and the point furthest away from the pole.

Find the distance between the pole and each of the following points on the curve:

a) The point on the tangent line which is perpendicular to the initial line.

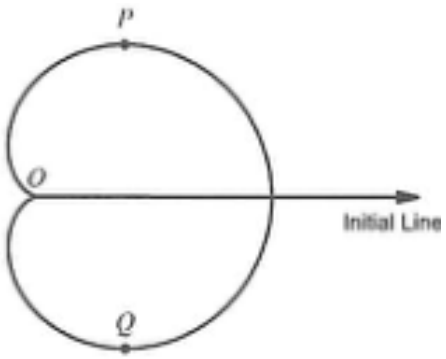
b) The point on the tangent line which is parallel to the initial line.

2.54 — Question 8. A curve is given by $r = \frac{1}{2}\sin 2\theta$, $0 \leq \theta < 2\pi$, and is bounded by the square $ABCD$ as shown in the diagram below.



Calculate the exact area of the square $ABCD$.

2.54 — Question 9. A polar curve is given by $r = a(1 + \cos\theta)$, $0 \leq \theta < 2\pi$, where a is a positive constant.

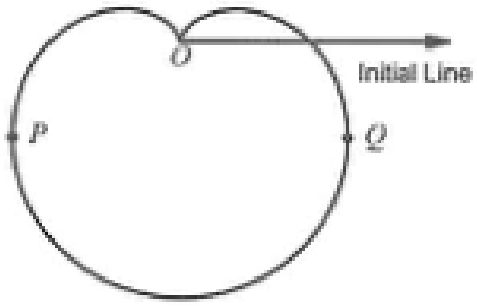


The points P and Q are where the tangent lines are parallel to the initial line. The distance between P and Q is known to be 12 units.

a) Determine the polar coordinates (r, θ) of the points P and Q.

b) Find the value of a giving your answer in exact form.

2.54 — Question 10. A polar curve is given by $r = a(1 - \sin\theta)$, $0 \leq \theta < 2\pi$, where a is a constant.



The points P and Q are where the tangent lines are perpendicular to the initial line. The distance between P and Q is known to be 5.

a) Determine the polar coordinates (r, θ) of the points P and Q.

b) Find the value of a giving your answer in exact form.

2.54 — Question 11. A polar curve is given by $r = k - 3\cos\theta$, $0 \leq \theta < 2\pi$, where $k > 3$. It is known that there are exactly four points on the curve such that the tangent is perpendicular to the initial line.

a) Show that the range of possible values for k is $3 < k < 6$.

b) Sketch $r = 5 - 3\cos\theta$ where $0 \leq \theta < 2\pi$. You should label any points at which it crosses the axes.

2.54 — Question 12. A polar curve is given by $r = k - 4\sin\theta$, $0 \leq \theta < 2\pi$, where $k > 4$. It is given that there are exactly four points on the curve such that the tangent is parallel to the initial line.

a) Show that the range of possible values for k is $4 < k < 8$.

b) Sketch $r = 6 - 4\sin\theta$ where $0 \leq \theta < 2\pi$. You should label any points at which it crosses the axes.

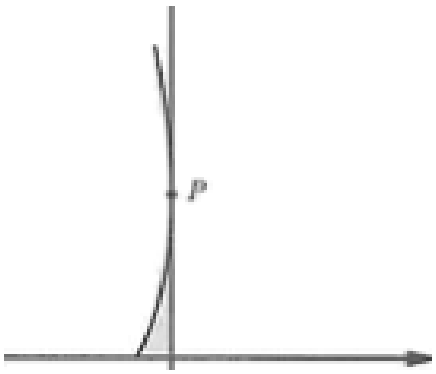
2.54 — Question 13. A polar curve C is given by $r = 4(1 - \sin\theta)$, $0 \leq \theta < \pi$.

a) Determine the horizontal line L which is a tangent to the curve.

b) Sketch the curve C and L on the same axes.

c) Find the area enclosed by the curve C and the line segment L, giving your answer in exact form.

2.54 — Question 14. A polar curve C is given by $r = 2 + \tan\theta$, $0 \leq \theta \leq \frac{\pi}{4}$.



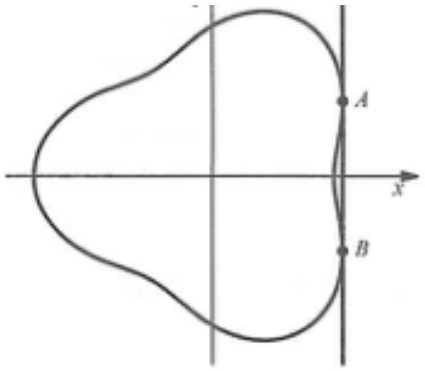
Point P is the point on the curve which has a tangent which is parallel to the initial line.

a) Show that the argument of the point P is given by $\sin\theta = \frac{1}{\sqrt{5}}$.

b) Find the equation of the tangent line in its Cartesian form.

c) Find the area which is bounded by the tangent line going through P, the curve and the initial line. Give your answer to three significant figures.

2.54 — Question 15. The diagram shows a sketch of the curve C given by $r = 6\sqrt{3} - 2\cos 3\theta$, $0 \leq \theta < 2\pi$.



The line L is a vertical tangent to the curve which passes through the points A and B.

a) Verify that the argument of point A is $\dfrac{\pi}{6}$.

b) Show that the area of the region bounded by C and L as shaded in the diagram is $\left(35\sqrt{3} - \dfrac{55\pi}{3}\right)$ units^2.

2.54 — Question 16. For each of the following curves, by using Calculus, find the value of θ for which the curve is furthest away from the pole and the corresponding distance from the pole:

a) $r = 3\cos\theta$ for $0 \leq \theta \leq \dfrac{\pi}{2}$

b) $r = 2 + \sin\theta$ for $0 \leq \theta \leq \pi$

c) $r = 5\cos 2\theta$ for $0 \leq \theta \leq \dfrac{\pi}{4}$

d) $r = \ln(1 + \sin\theta)$ for $0 \leq \theta < \pi$

2.54 — Question 17. A curve C is given by $r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right)$, $\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$. The point P lies on the curve at the furthest point at angle ϕ.

a) Explain why C consists of a single loop.

b) Show that ϕ satisfies the equation $\phi = \ln\left(\frac{1+4\tan 2\phi}{1-4\tan 2\phi}\right)$.

c) Given that $\phi = 1.65488$ correct to six significant figures, find the distance from P to the pole, giving your answer to four significant figures.

2.54 — Question 18. Let $r = r(\theta)$ be a differentiable function. Given that a curve may be defined parametrically by $x = r\cos\theta$ and $y = r\sin\theta$, show that

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}.$$

2.54 — Question 19. For each of the following polar curves, find the gradient of the tangent line at the given value of θ. Give your answers in exact form.

a) $r = 2\theta$ for $\theta = \dfrac{\pi}{2}$

b) $r = 2\cos\theta$ for $\theta = \dfrac{\pi}{6}$

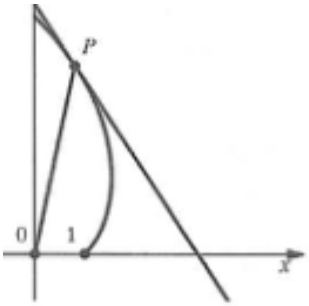
c) $r = 1 + 2\sin\theta$ for $\theta = \dfrac{3\pi}{4}$

2.54 — Question 20. A curve C is given by $r = \theta\sin\theta$ for $0 \leq \theta < \pi$.

a) Show that $\dfrac{dy}{dx} = \dfrac{2\theta\sin 2\theta + 2\sin^2\theta}{2\theta\cos 2\theta + \sin 2\theta}$.

b) Hence, show that this curve consists of a single loop.

2.54 — Question 21. A curve C has a polar equation $r = e^{k\theta}$ for $0 < \theta \leq \frac{\pi}{2}$, where k is a positive constant.



at point P on curve C is L
radient of the tangent line L

A tangent line at point P on curve C is L.

a) Find the gradient of the tangent line L.

b) Show that the angle, ϕ, between the line OP and L satisfies $\tan\phi = \frac{1}{k}$.

2.54 — Question 22. Show that for the curve $r = \sin 2\theta$, the gradients $\frac{dy}{dx}$ at points equidistant from $\theta = \frac{\pi}{4}$ are symmetrically related.